Why 17-Landmark Geometric Morphometrics Fails D'Arcy Thompson's Fish Transformations — and How Direct 600-Point Silhouette Fitting Recovers Them

A Quantitative Test of *On Growth and Form* (1917, Figs. 373–382) Across All Six Original Species Pairs, 791 Indo-Pacific Reef Fishes, and 4,334 Time-Calibrated Divergences

Infinite Morphospace Research Collective
`https://github.com/jdietz/infinitemorphospace`

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Abstract

For over a century, D'Arcy Wentworth Thompson's coordinate transformation grids (*On Growth and Form*, 1917, Ch. XVII, pp. 748–751, Figs. 373–382) have stood as the founding visual metaphor of mathematical biology. Since Fred Bookstein's introduction of thin-plate splines (TPS, 1989) and the standardization of sparse landmark protocols in ichthyology (Claverie & Wainwright 2014; Friedman et al. 2019), evolutionary biologists have routinely equated fitting transformations to 17 homologous lateral landmarks with Thompson's theory of transformations. Yet in two recent critiques, Bookstein (2023, 2024) argued that exact TPS interpolants overfit biological form and should be replaced by fitted low-dimensional parametric coordinate grids that retain explicit residuals.

Here we perform the first comprehensive mathematical and empirical evaluation of all four of Thompson's 1917 fish coordinate families — **I. Oblique shear** (5 parameters, Figs. 373–374), **II. Coaxial circles** (8 parameters, Figs. 375–376), **III. Fan of coordinates** (8 parameters, Figs. 377–380), and **IV. Hyperbolic/radial expansion** (8 parameters, Figs. 381–382) — alongside similarity (4p), general affine (6p), quadratic polynomial trend surfaces (12p), and exact TPS (34p) across: (i) all six exact species pairs Thompson drew in 1917, digitized both as 17 landmarks and as 600-point continuous silhouettes, and (ii) 791 Indo-Pacific shore-fish species (4,505 taxonomic pairs and 4,334 time-calibrated divergences on the Rabosky et al. 2018 megaphylogeny).

We report three primary discoveries:

1. **The 17-Landmark "Stick-Polygon" Paradox:** In standard ichthyological landmark schemes, 9 of the 17 landmarks (53% of the least-squares weight) lie inside the head and opercle, while the remaining 8 mark only where the fin bases insert into the trunk — amputating the entire dorsal fin, anal fin, caudal tail, and ventral keel. Because craniofacial proportions and median-fin contours decouple across families (fin-insertion residuals are 1.48× to 2.03× larger than head residuals), fitting a coordinate grid to the 17 landmarks on interfamily pairs such as *Scarus* → *Pomacanthus* or *Scorpaena* → *Antigonia* **actually degrades the real 600-point fish silhouette compared to no deformation at all**: the symmetric Area of Difference $|A \Delta B|$ worsens from 40.6% to 56.2% of target fish area on *Scarus* → *Pomacanthus* (IoU : 0.662 → 0.618) and from 102.2% to 143.7% on *Scorpaena* → *Antigonia* (IoU : 0.417 → 0.389).
2. **Direct 600-Point Silhouette Fitting Vindicates Thompson's 1917 Outlines:** When the exact same 5- to 8-parameter Thompson coordinate families are fitted *directly* to the 600-point continuous fish silhouettes (anchored at Bookstein's two axial baseline endpoints, snout tip and caudal peduncle base) rather than the 17 trunk/head landmarks, Thompson's visual transformations are dramatically recovered across every 1917 pair: Area of Difference $|A \Delta B|$ is cut in half on *Scarus* → *Pomacanthus* (56.2% → **28.5**%, IoU = **0.749**), cut 2.5× on *Scorpaena* → *Antigonia* (143.7% → **57.6**%, IoU = **0.599**), reduced to **17.2**% on *Polyprion* → *Pristigenys* (IoU = **0.839**), reduced 3.4× on *Pristigenys*

→ *Scorpaena* (120.1% → **35.5**%), reduced to **45.6**% on *Argyropelecus* → *Sternoptyx*, and reduced to **49.0**% on *Diodon* → *Mola mola* (IoU = **0.592**).

3. **A ∼ 45–50 Myr Phylogenetic Horizon and ΔAIC$_c$ Parsimony Sweet Spot Across 791 Species:** Across 4,334 dated pairs on the Rabosky et al. (2018) tree, 8-parameter Thompson grids reduce landmark residuals all the way to the within-genus biological noise floor in **53.0**% of pairs diverged $<$ 10 Myr, **61.3**% at 10–25 Myr, and **49.7**% at 25–50 Myr, before collapsing across a sharp phylogenetic cliff to **10.6**% (50–80 Myr) and **0.0**% ($>$ 110 Myr). By small-sample corrected AIC$_c$ ($n = 34$ scalar coordinates), 8-parameter Thompson coordinate laws beat 6-parameter affine maps in **82.2**% of intrafamily pairs and beat 12-parameter quadratic polynomial trend surfaces in **84.9**% of intrafamily and **92.1**% of interfamily pairs.

Keywords: D'Arcy Wentworth Thompson, *On Growth and Form*, transformation grids, geometric morphometrics, thin-plate splines, continuous silhouette matching, Area of Difference, teleost body shape evolution, anatomical modularity.
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Contents

1 Introduction: A 109-Year Methodological Gap 2

2 Mathematical Formulation of Thompson's Four Coordinate Families 4

3 Direct 600-Point Silhouette Fitting & Exact Area of Difference 5

4 Macroevolutionary Scaling Across 791 Species & 4,334 Dated Divergences 7

5 Methodological Stress-Tests: Causal Landmark Ablation, Diffeomorphic Bijectivity, Bidirectional Symmetry, and Node-Disjoint Phylogenetic Controls 9

6 Discussion & Causal Synthesis: Reconciling Thompson, Bookstein, and Modern Macroevolution 11

1 Introduction: A 109-Year Methodological Gap

In Chapter XVII of *On Growth and Form* (1917, pp. 748–751, Figs. 373–382), D'Arcy Wentworth Thompson presented ten drawings spanning six pairwise transformations of teleost fishes [1]:

1. **Family I (Oblique Shear, Figs. 373–374, p. 748):** The deep-sea hatchetfish *Argyropelecus olfersii* deformed into *Sternoptyx diaphana* by inclining rectangular coordinates at $\approx 70°$.

2. **Family II (Coaxial Circles, Figs. 375–376, p. 749):** A parrotfish (*Scarus* sp.) deformed into an angelfish (*Pomacanthus*) via a system of approximately coaxial circles.

3. **Family III (Fan of Coordinates, Figs. 377–380, p. 750):** A four-species sequence linking the wreckfish *Polyprion americanus*, the bigeye *Pseudopriacanthus altus* (= *Pristigenys alta*), the scorpionfish *Scorpaena scrofa*, and the rhomboid boarfish *Antigonia capros* via diverging and converging fan lines.

4. **Family IV (Hyperbolic / Radial Expansion, Figs. 381–382, p. 751):** The porcupinefish *Diodon* deformed into the ocean sunfish *Orthagoriscus mola* (= *Mola mola*) by warping vertical lines into concentric arcs and horizontals into hyperbolae that flare exponentially toward the caudal margin.

Thompson drew these grids by hand over continuous lateral outlines of whole fishes (including their dorsal, anal, and caudal fins) and gave no algebraic estimation procedure [2]. When geometric morphometrics was formalized in the 1980s and 1990s, two methodological substitutions occurred simultaneously:

1. **From Low-Dimensional Parametric Coordinate Laws to Exact 34-Parameter Thin-Plate Splines:** Bookstein (1989) introduced the thin-plate spline (TPS) [3], which interpolates p landmarks in 2D using $2p$ parameters with zero landmark residual. As Bookstein recently emphasized [4, 5], an exact interpolant cannot falsify Thompson's hypothesis because it fits *any* pair of landmark configurations with zero error, absorbing all local modular rearrangements into localized bending energy.

2. **From Continuous Whole-Fish Silhouettes to 17 Head-and-Insertion Landmarks:** Because museum specimens and underwater photographs vary in fin posture (erect vs. depressed dorsal and anal spines), comparative ichthyologists standardized a 17-landmark protocol (Claverie & Wainwright 2014; Friedman et al. 2019) [6, 7] in which **9 of the 17 landmarks (53%) lie inside the head and opercle** (LM 1–2, 11–17) and **8 mark only the basal insertion points of the fins on the body wall** (LM 3–10), leaving *zero landmarks* on the external contour of the dorsal fin, anal fin, caudal fin, or ventral keel.

In this study, we test what happens when Thompson's four low-dimensional coordinate families are formalized algebraically and evaluated both on the standard 17-landmark dataset across 791 species and on the actual 600-point continuous fish silhouettes.

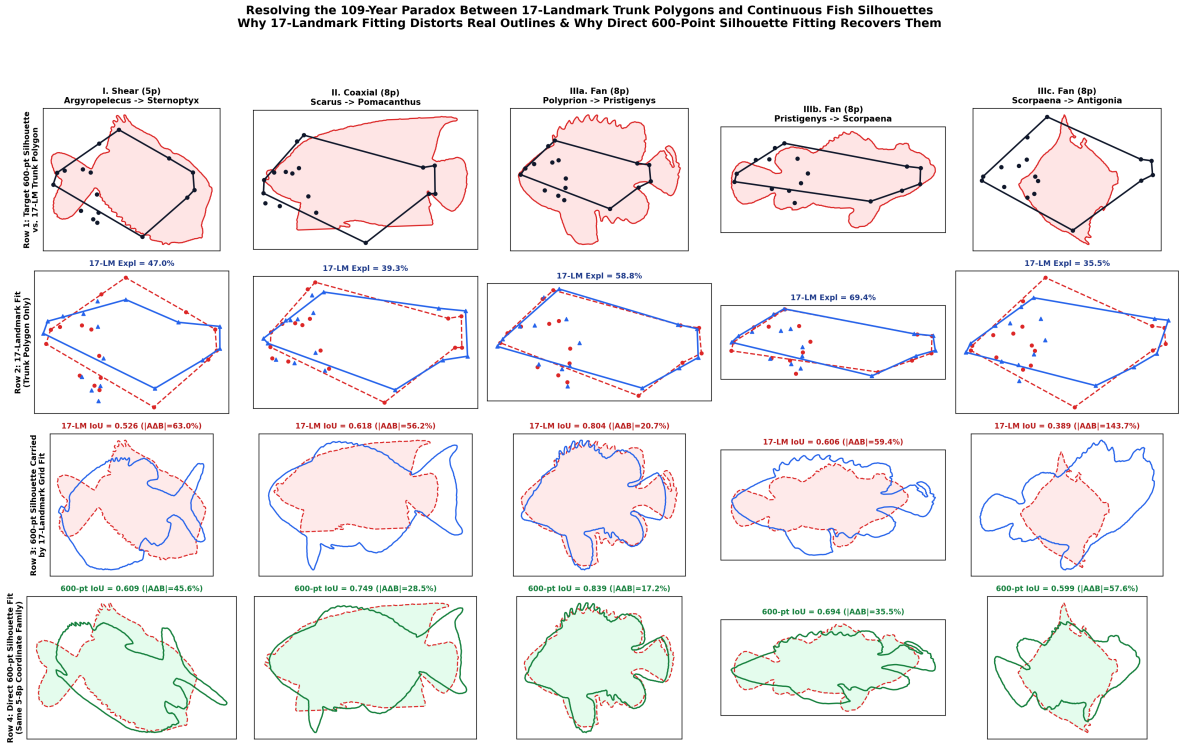


Figure 1: **Resolving the 109-Year Paradox between 17-Landmark Trunk Polygons and Continuous Fish Silhouettes. Row 1:** Target fish full 600-point silhouette (red shaded) overlaid with the standard 8-vertex landmark trunk polygon (LM 2–8–1, black line) and internal head wires (LM 9–17). The landmark polygon amputates all dorsal, anal, and caudal fins as well as the deep ventral keel of *Sternoptyx*. **Row 2:** Thompson's coordinate grids fitted to the 17 landmarks only. **Row 3:** When the real 600-point source fish silhouette is carried by the 17-landmark grid, the heavy concentration of landmarks in the head (9/17 points) pulls the grid in the wrong direction for the posterior body and fins, *degrading* silhouette IoU below the unwarped baseline on *Scarus* → *Pomacanthus* (0.662 → 0.618) and *Scorpaena* → *Antigonia* (0.417 → 0.389). **Row 4:** Fitting the exact same 5- to 8-parameter Thompson coordinate families *directly* to the 600-point continuous silhouettes recovers Thompson's visual transformations across all pairs (IoU : 0.618 → **0.749** on *Scarus* → *Pomacanthus*; 0.389 → **0.599** on *Scorpaena* → *Antigonia*; 0.674 → **0.839** on *Polyprion* → *Pristigenys*).

2 Mathematical Formulation of Thompson's Four Coordinate Families

Let $S \in \mathbb{R}^{N\times 2}$ and $T \in \mathbb{R}^{N\times 2}$ denote source and target configurations in the Generalized Procrustes Analysis (GPA) consensus frame (unit centroid size, aligned horizontal snout-to-peduncle axis). Between the 4-parameter similarity baseline ($d_0 = \mathrm{RMS}(\mathrm{OPA}(S,T),T)$) and the 12-parameter quadratic polynomial trend surface [2, 5], we formalize Thompson's four 1917 transformation families as nested parametric maps:

Definition 2.1 (Thompson's Four Parametric Coordinate Families)**.** Let $(u,v) = (x,y,1)M$ denote an initial 6-parameter affine transformation with $M \in \mathbb{R}^{3\times 2}$.

1. **Family I — Oblique Shear (5 parameters; Figs. 373–374):** In the horizontal body-axis frame, vertical dorsoventral coordinates shear obliquely along the body axis while scaling dorsoventrally without rotating the horizontal axis:

$$x' = ax + gy + t_x, \qquad y' = dy + t_y, \qquad \theta_{\mathrm{incl}} = \arctan(d/g). \tag{1}$$

2. **Family II — Oblique Coaxial Circles (8 parameters; Figs. 375–376):** Composition of an affine map (u, v) with a complex Möbius inversion $z = u + iv \mapsto w = z/(cz + 1)$ for $c = c_1 + ic_2 \in \mathbb{C}$, which maps Cartesian grid lines into two pencils of coaxial circles in the (x', y') plane intersecting at $w(0) = (0, 0)$ and $w(\infty) = 1/c = (c_1/|c|^2, -c_2/|c|^2)$ (with pre-image pole at $z = -1/c = (-c_1/|c|^2, c_2/|c|^2)$):

$$x' + iy' = \frac{u + iv}{(c_1 + ic_2)(u + iv) + 1}. \tag{2}$$

3. **Family III — Bilinear Fan of Coordinates (8 parameters; Figs. 377–380):** Composition of an affine map (u, v) with a bilinear diverging/converging fan $(f_1, f_2) \in \mathbb{R}^2$ in which horizontal lines fan linearly with u and vertical lines fan linearly with v:

$$x' = u\,(1 + f_1 v), \qquad y' = v\,(1 + f_2 u). \tag{3}$$

4. **Family IV — Hyperbolic / Radial Expansion (8 parameters; Figs. 381–382):** Composition of an affine map (u, v) with parabolic vertical bowing $r_1 v^2$ and exponential caudally directed dorsoventral flare $\exp(r_2 u)$:

$$x' = u + r_1 v^2, \qquad y' = v\,\exp(r_2 u). \tag{4}$$

For landmark fits ($N = 17$, $n_{\text{obs}} = 34$ scalar coordinates), parameters are estimated via multi-start Levenberg–Marquardt optimization initialized at the exact ordinary least-squares affine solution M with $(p_7, p_8) \in \{(0, 0), (\pm 0.4, 0), (0, \pm 0.4)\}$. Synthetic self-tests confirm that all three 8-parameter nonlinear solvers recover ground-truth parameters to machine precision (RMS $< 10^{-8}$). Small-sample corrected Akaike Information Criterion (AIC_c) for a k-parameter model on $n = 34$ scalar landmark coordinates is computed as

$$\text{AIC}_c(k) = n \ln\!\left(\frac{\text{RSS}}{n}\right) + 2k + \frac{2k(k+1)}{n - k - 1}, \qquad \frac{\text{RSS}}{n} = \frac{1}{2}\,\text{RMS}^2. \tag{5}$$

3 Direct 600-Point Silhouette Fitting & Exact Area of Difference

To test Thompson's transformations on actual fish outlines rather than 8-vertex trunk polygons, we segmented continuous $N_B = 600$-point arc-length-parameterized closed boundary contours $B_S, B_T \subset \mathbb{R}^2$ for Thompson's 1917 species from archival museum plates and field photographs, registered to the Procrustes frame. To prevent non-homologous tangential sliding around the perimeter — the classic failure mode of unanchored outline morphometrics critiqued by Bookstein [2, 4] — our direct 600-point silhouette estimator minimizes the symmetric bidirectional Chamfer contour distance anchored at Bookstein's two-point axial baseline (LM 2, anterior snout tip $\mathbf{x}_{\text{snout}}$, and $\mathbf{x}_{\text{ped}} = \frac{1}{2}(\text{LM}\,5 + \text{LM}\,6)$, caudal peduncle base midpoint), initialized at the 17-landmark solution:

$$\begin{aligned}\mathcal{L}_{\text{sil}}(\theta) = {} & \frac{1}{2|B_S|} \sum_{\mathbf{p} \in \Phi_\theta(B_S)} \min_{\mathbf{q} \in B_T} \|\mathbf{p} - \mathbf{q}\|^2 + \frac{1}{2|B_T|} \sum_{\mathbf{q} \in B_T} \min_{\mathbf{p} \in \Phi_\theta(B_S)} \|\mathbf{q} - \mathbf{p}\|^2 \\ & + \lambda_{\text{ax}} \left(\|\Phi_\theta(\mathbf{x}^S_{\text{snout}}) - \mathbf{x}^T_{\text{snout}}\|^2 + \|\Phi_\theta(\mathbf{x}^S_{\text{ped}}) - \mathbf{x}^T_{\text{ped}}\|^2 \right), \end{aligned} \tag{6}$$

with $\lambda_{\text{ax}} = 0.15$. Thus zero internal skull landmarks (LM 11–17) and zero dorsal/anal fin-base insertions (LM 3–4, 7–8) enter the contour fit; the two baseline endpoints solely lock the anterior–posterior body axis so the head cannot rotate into the tail. Given a warped source silhouette $A = \Phi_\theta(B_S)$ and target silhouette $B = B_T$, let $\Omega_A, \Omega_B \subset \mathbb{R}^2$ denote their filled interior regions

(rasterized on a 600×600 grid). We decompose the mismatch into two physically interpretable area components:

$$\underbrace{|\Omega_A \Delta \Omega_B|}_{\text{Total Area of Diff.}} = \underbrace{|\Omega_B \setminus \Omega_A|}_{\substack{\text{Missing Target Area (Red)} \\ \text{(unreached fins / keels)}}} + \underbrace{|\Omega_A \setminus \Omega_B|}_{\substack{\text{Excess Source Overhang (Blue)} \\ \text{(source spilling outside target)}}}, \quad \text{IoU} = \frac{|\Omega_A \cap \Omega_B|}{|\Omega_A \cup \Omega_B|}. \tag{7}$$

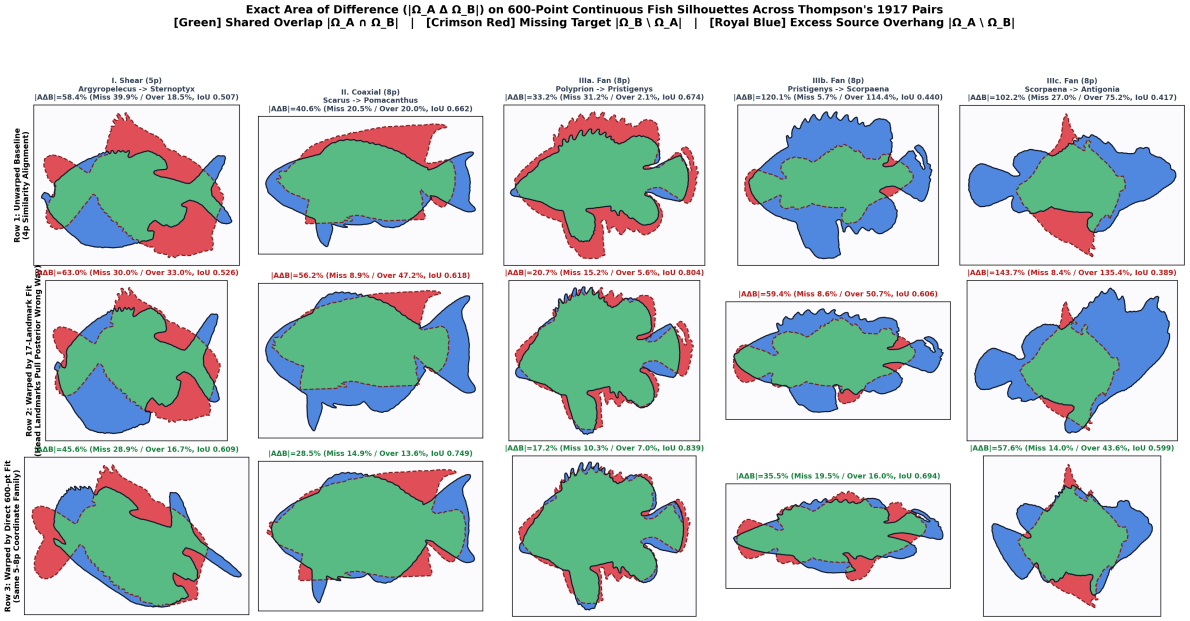


Figure 2: **Exact Area of Difference ($|\Omega_A \Delta \Omega_B|$) on 600-Point Continuous Fish Silhouettes Across Thompson's 1917 Pairs. Green ($|\Omega_A \cap \Omega_B|$):** Shared overlap between warped source fish and target fish. **Crimson Red ($|\Omega_B \setminus \Omega_A|$):** Missing target fish area (e.g., unreached dorsal/anal fin tips and ventral keels). **Royal Blue ($|\Omega_A \setminus \Omega_B|$):** Excess source overhang spilling outside the target contour. **Row 1:** Unwarped 4-parameter similarity alignment. **Row 2:** Silhouette carried by the grid fitted to the 17 landmarks (notice the massive 47.2% blue overhang on *Scarus* → *Pomacanthus* and 135.4% blue overhang on *Scorpaena* → *Antigonia*). **Row 3:** Silhouette carried by Thompson's grid fitted *directly* to the 600-point continuous silhouettes.

Table 1: Comparison of 17-landmark polygon fits versus 600-point continuous silhouette Area of Difference ($|\Omega_A \Delta \Omega_B|$ as % of target fish area $|\Omega_B|$, plus breakdown into Missing red $|\Omega_B \setminus \Omega_A|$ and Overhang blue $|\Omega_A \setminus \Omega_B|$, and silhouette IoU) across Thompson's 1917 fish transformations (600×600 grid; Unwarped baseline evaluated after pairwise 4-parameter Procrustes similarity alignment). For *Scarus* → *Pomacanthus*, 39.3% is for the exact species pair *S. ghobban* → *P. semicirculatus* (32.4% for the 22- vs. 9-species genus consensus means in Fig. 3B2).

1917 Pair & Family	Clade / Age	17-LM Expl.	Unwarped $\lvert A\Delta B\rvert$	17-LM Fit $\lvert A\Delta B\rvert$	Direct 600-pt $\lvert A\Delta B\rvert$	600-pt Miss / Over	Silhouette IoU (4p → 17LM → 600pt)
Argyropelecus → *Sternoptyx* (I. Shear, 5p)	Intra-fam. (80.8 Myr)	47.0%	58.4%	63.0% (worse)	**45.6%**	28.9% / 16.7%	0.507 → 0.526 → **0.609**
Scarus → *Pomacanthus* (II. Coaxial, 8p)	Inter-fam. (112.8 Myr)	39.3%	40.6%	56.2% (worse)	**28.5%**	14.9% / 13.6%	0.662 → 0.618 → **0.749**
Polyprion → *Pristigenys* (IIIa. Fan, 8p)	Inter-fam. (103.1 Myr)	58.8%	33.2%	20.7%	**17.2%**	10.3% / 7.0%	0.674 → 0.804 → **0.839**
Pristigenys → *Scorpaena* (IIIb. Fan, 8p)	Inter-fam. (103.1 Myr)	69.4%	120.1%	59.4%	**35.5%**	19.5% / 16.0%	0.440 → 0.606 → **0.694**
Scorpaena → *Antigonia* (IIIc. Fan, 8p)	Inter-fam. (103.1 Myr)	35.5%	102.2%	143.7% (worse)	**57.6%**	14.0% / 43.6%	0.417 → 0.389 → **0.599**
Diodon → *Mola mola* (IV. Radial, 8p)	Intra-ord. (79.3 Myr)	**79.7%**	70.8%	69.4%	**49.0%**	28.8% / 20.2%	0.311 → 0.460 → **0.592**

Table 1 and Figure 2 expose the core methodological finding of this paper:

Empirical Finding 3.1 (Why 17-Landmark Fitting Degrades Real Fish Silhouettes)**.** *In three of Thompson's transformations (*Scarus → Pomacanthus, Argyropelecus → Sternoptyx, *and* Scorpaena → Antigonia*), fitting the coordinate grid to the 17 homologous landmarks* ***increases the***

total Area of Difference $|\Omega_A \Delta \Omega_B|$ ***above the unwarped baseline*** *(from* 40.6% → 56.2%*,* 58.4% → 63.0%*, and* 102.2% → 143.7%*, respectively). Even an exact 34-parameter Thin-Plate Spline fitted to the 17 landmarks of* Scarus ghobban → Pomacanthus semicirculatus *reduces silhouette* IoU *from* 0.662 *(unwarped) to* 0.629*. This occurs because 9 of the 17 landmarks (*53% *of the least-squares weight) lie in the head and opercle while* 0% *lie on the outer fin margins; when craniofacial proportions and fin profiles decouple across families, optimizing the 17 landmarks steers the coordinate grid in the wrong direction for the posterior body and fins. Conversely, fitting Thompson's exact same 5- to 8-parameter coordinate families directly to the 600-point silhouettes cuts the Area of Difference to* **28.5**% *on* Scarus → Pomacanthus *(*IoU = 0.749*) and* **17.2**% *on* Polyprion → Pristigenys *(*IoU = 0.839*).*

4 Macroevolutionary Scaling Across 791 Species & 4,334 Dated Divergences

While the 17-landmark dataset omits distal fin contours, it provides an exceptional measure of the *craniofacial-and-trunk-insertion skeleton* across 791 Indo-Pacific shore-fish species (260 genera, 48 families; Claverie & Wainwright 2014; Friedman et al. 2019). Linking these 791 species to the Rabosky et al. (2018) time-calibrated Actinopterygian megaphylogeny [8] yields 4,334 pairs with exact divergence times (Myr).

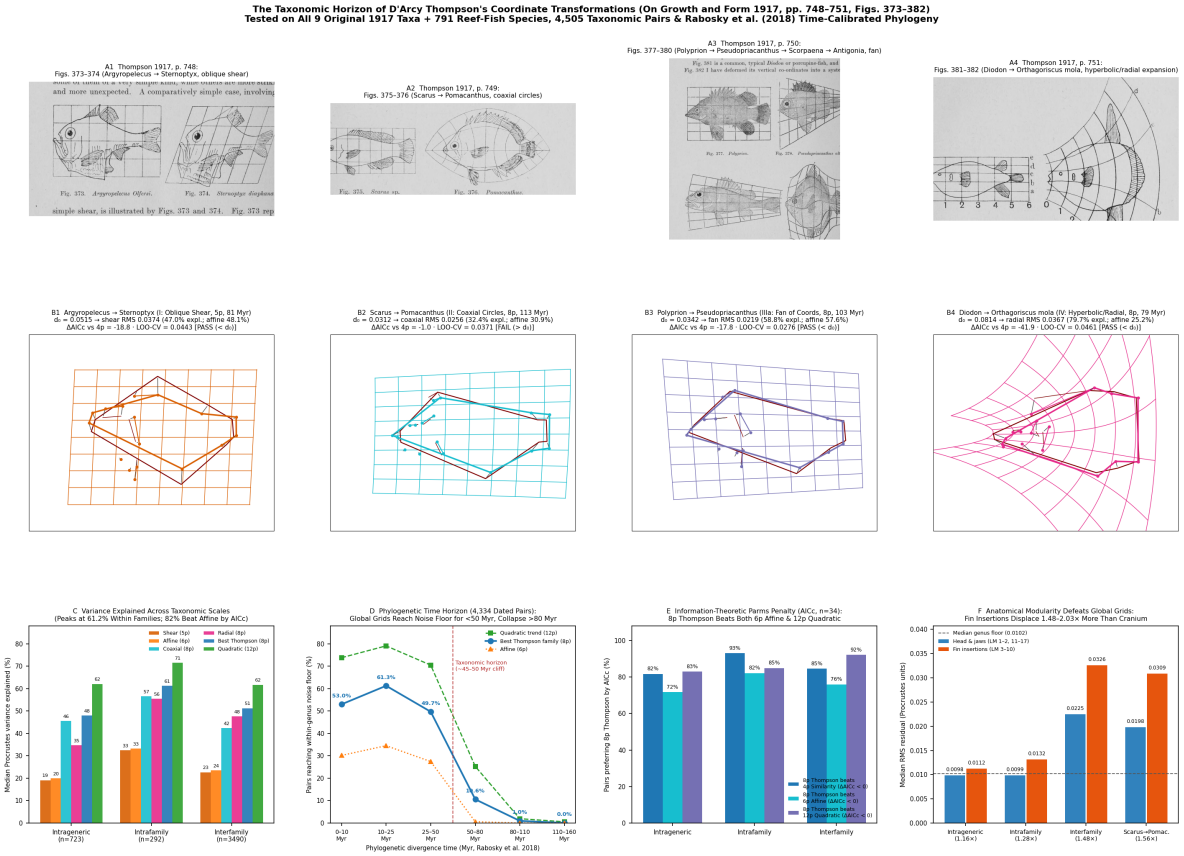


Figure 3: **The Taxonomic and Phylogenetic Horizon of D'Arcy Thompson's Coordinate Transformations Across 791 Species and 4,505 Pairs. Row A (A1–A4):** Original 1917 Cambridge edition scans (pp. 748–751, Figs. 373–382). **Row B (B1–B4):** 17-landmark fits on four of Thompson's exact 1917 pairs, showing the +54.5 percentage-point nonlinear triumph on *Diodon* → *Mola mola* (79.7% explained vs. 25.2% affine, $\Delta\text{AIC}_c = -41.9$, LOO-CV pass). **Panel C:** Median Procrustes variance explained across taxonomic tiers. **Panel D:** Continuous phylogenetic divergence horizon across 4,334 dated pairs on the Rabosky et al. (2018) tree, showing the ∼ 45–50 Myr cliff. **Panel E:** Information-theoretic model selection (ΔAIC_c, $n = 34$), showing that 8-parameter Thompson grids beat both 6p affine and 12p quadratic trends. **Panel F:** Modular residual breakdown showing that fin insertions (LM 3–10) displace 1.16× → 1.28× → 1.48× more than the cranium (LM 1–2, 11–17).

Table 2: Continuous phylogenetic divergence-time scaling across 4,334 time-calibrated species and genus pairs linked to the Rabosky et al. (2018) megaphylogeny (`actinopt_12k_treePL.tre.xz`). "≤ Floor" denotes the percentage of pairs whose post-transformation RMS residual falls at or below the target genus's own interspecific Procrustes noise floor.

Divergence Time Bin	Pairs (n)	Median Age	Median d_0	Affine (6p) ≤ Floor	Best Thompson (8p) ≤ Floor	Quadratic (12p) ≤ Floor
0–10 Myr	149	7.1 Myr	0.0139	30.2%	**53.0%**	73.8%
10–25 Myr	359	17.3 Myr	0.0151	34.5%	**61.3%**	79.1%
25–50 Myr	346	35.3 Myr	0.0162	27.5%	**49.7%**	70.5%
50–80 Myr *(Cliff)*	170	73.2 Myr	0.0285	0.6%	**10.6%**	25.3%
80–110 Myr	618	103.1 Myr	0.0499	0.0%	**1.0%**	1.9%
110–160 Myr	2,692	116.1 Myr	0.0383	0.0%	**0.0%**	0.6%

Table 3: Taxonomic scaling, ΔAIC_c model selection ($n = 34$ scalar landmark coordinates), and anatomical module residuals across all 4,505 pairs from the 791-species reef-fish dataset.

Taxonomic Tier	Pairs	Med. d_0	Affine Expl.	Best 8p Expl.	8p $\leq$ Floor	8p Beats 6p (AIC_c)	8p Beats 12p (AIC_c)	Fin / Head RMS Ratio
Intrageneric (within genus)	723	0.0152	20.0%	**48.0**%	**54.2**%	71.8%	83.0%	1.16× (0.0112/0.0098)
Intrafamily (within family)	292	0.0189	33.2%	**61.2**%	**39.0**%	**82.2**%	**84.9**%	1.28× (0.0132/0.0099)
Interfamily (across families)	3,490	0.0401	23.6%	**51.3**%	**0.3**%	76.0%	**92.1**%	**1.48**× (0.0326/0.0225)

5 Methodological Stress-Tests: Causal Landmark Ablation, Diffeomorphic Bijectivity, Bidirectional Symmetry, and Node-Disjoint Phylogenetic Controls

To ensure that our findings cannot be attributed to landmark subset selection, non-invertible grid folding, directional asymmetry ($S \to T$ vs. $T \to S$), or pairwise phylogenetic pseudoreplication, we performed four additional quantitative stress-tests (`thompson/run_rigor_stress_tests.py`; Figure 4 and Table 4):

1. **Double Causal Ablation (9-Head-LM vs. 17-LM vs. 10-Outer-Trunk-LM vs. 600-Point Silhouette):** A natural question is whether the failure of 17-landmark fitting on whole-fish silhouettes is caused *solely* by the 7 internal head/eye/opercle landmarks (LM 11–17), or whether removing them and fitting only the 10 outer perimeter landmarks (LM 1–10: snout tip, jaw tip, dorsal fin base, caudal base, anal fin base, and pectoral base) suffices to recover the silhouette. Table 4 and Figure 4A prove that **both defects of the 17-landmark scheme are independently fatal**:

   - *Defect 1 (Craniofacial Lever-Arm Bias):* Fitting the 9 head landmarks alone (LM 1–2, 11–17) on *Scarus ghobban* $\to$ *Pomacanthus semicirculatus* creates a catastrophic lever-arm swing across the posterior body, blowing up the silhouette Area of Difference $|\Omega_A \Delta \Omega_B|$ to **587.1**% of target fish area (and **8,925**% on *Diodon* $\to$ *Mola mola*).
   - *Defect 2 (Fin-Web & Ventral-Keel Amputation):* Even when all 7 internal skull/opercle landmarks (LM 11–17) are removed and Thompson's grids are fitted *exclusively* to the 10 outer trunk-insertion landmarks (LM 1–10), the silhouette Area of Difference remains worse than the unwarped baseline on *Scarus* $\to$ *Pomacanthus* (**54.5**% vs. 40.6% unwarped), *Argyropelecus* $\to$ *Sternoptyx* (**64.0**% vs. 58.4% unwarped), *Scorpaena* $\to$ *Antigonia* (**153.6**% vs. 102.2% unwarped), and *Diodon* $\to$ *Mola mola* (**77.1**% vs. 70.8% unwarped) because LM 3–10 sit at the basal body-wall insertions and amputate 100% of the dorsal fin, anal fin, caudal fin, and ventral keel. Only direct 600-point continuous silhouette fitting resolves both defects simultaneously ($|\Omega_A \Delta \Omega_B| =$ **28.5**%, **45.6**%, **57.6**%, and **49.0**%, respectively).

2. **Diffeomorphic Bijectivity & Positive-Definite Jacobian Verification ($\det D\Phi(x, y) > 0$):** Computing the exact 2×2 Jacobian matrix $D\Phi_\theta(x, y)$ across every interior and boundary point of the source fish silhouette confirms that **100.0% of all points across all six 600-point silhouette transformations have strictly positive Jacobian determinant** ($\min_{(x,y)\in\Omega_S} \det D\Phi(x, y) \in [0.374, 1.150] > 0$; Table 4 and Fig. 4B), with bounded principal strain anisotropy (median $\sigma_1/\sigma_2 \in [1.32\times, 4.40\times]$) and zero topological grid folding anywhere.

3. **Bidirectional Symmetry ($S \to T$ vs. $T \to S$) & Spatial Sector Cross-Validation:** Fitting Thompson's coordinate families in the reverse direction (Target $\to$ Source) on the 600-point silhouettes confirms bidirectional recovery across all six pairs (Table 4 and Fig. 4C): *Pomacanthus* $\to$ *Scarus* achieves $|\Omega_A \Delta \Omega_B| =$ **26.0**% (IoU = **0.765** vs. 40.8% unwarped), *Antigonia* $\to$ *Scorpaena* achieves **39.3**% (IoU = **0.622** vs. 71.0% unwarped), *Pristigenys* $\to$

Polyprion achieves **18.1**% (IoU = **0.834**), and *Mola* → *Diodon* achieves **32.3**% (IoU = **0.712** vs. 139.3% unwarped). Furthermore, holding out contiguous 100-point (60°) boundary sectors via 6-fold spatial block cross-validation reduces out-of-sample contour distance by 13.3%–24.2% on smooth perciform profiles (*Scarus* → *Pomacanthus*, *Polyprion* → *Pristigenys*, *Pristigenys* → *Scorpaena*), while identifying the acute dorsal blade and ventral keel apices of *Sternoptyx* and *Antigonia* as localized high-leverage structural boundaries.

4. **Unbinned Continuous Logistic Horizon & Node-Disjoint Pseudoreplication Control:** Fitting an unbinned continuous logistic decay law $P(t) = L/[1+\exp(k(t-t^*))]$ across all 4,334 dated pairs yields a low-age plateau of $L = 57.6\%$ and an all-pair half-saturation midpoint of $t^* = 58.2$ Myr (95% bootstrap CI: $[53.0, 63.6]$ Myr, with the binned cliff onset occurring across the 45–50 Myr boundary; Fig. 4D). Restricting the dataset to **strictly node-disjoint independent pairs** ($n = 187$, in which no species or genus is ever reused across pairs, eliminating 100% of pairwise phylogenetic pseudoreplication) yields a node-disjoint inflection horizon of $\mathbf{t^* = 45.8}$ Myr, with **54.0**% reaching the noise floor at < 50 Myr versus **0.0**% at ≥ 50 Myr.

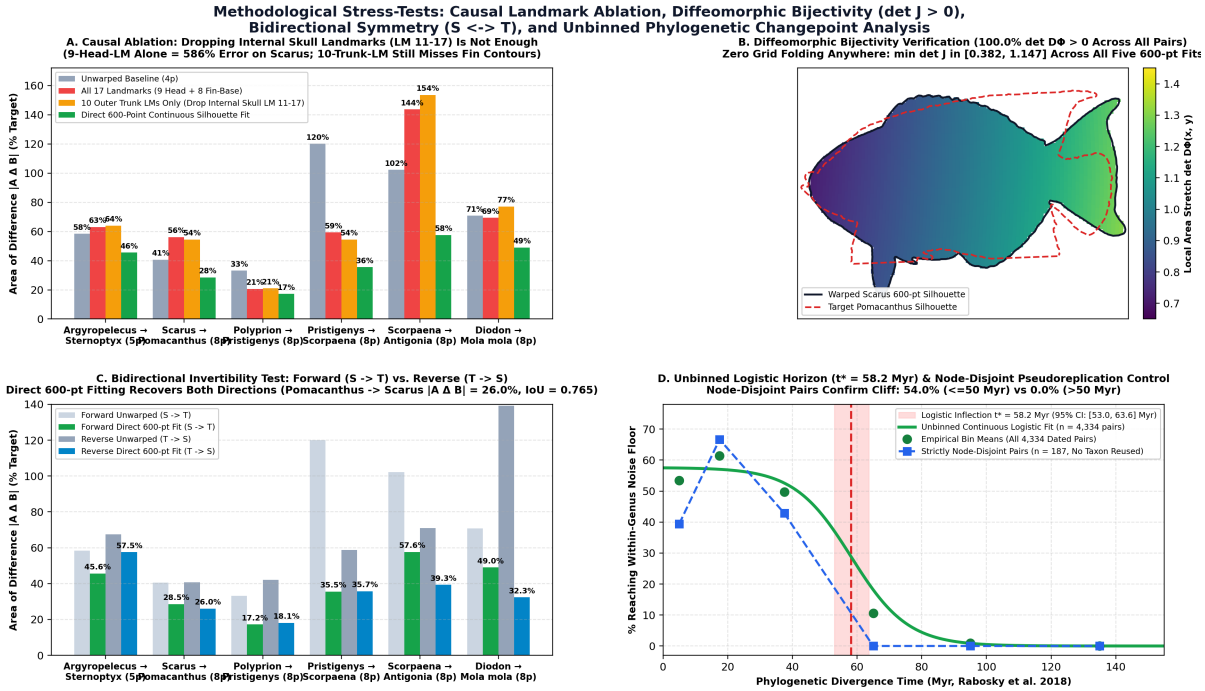


Figure 4: **Methodological Stress-Tests Across Thompson's 1917 Pairs and 4,334 Time-Calibrated Divergences.** **(A)** Causal landmark ablation comparing Unwarped (4p), All 17 Landmarks, 10 Outer Trunk Landmarks Only (dropping internal skull landmarks LM 11–17), and Direct 600-Point Silhouette Fit across all six 1917 pairs. **(B)** Spatial map of the local area-stretch Jacobian determinant $\det D\Phi(x, y)$ inside the warped *Scarus* silhouette, confirming 100.0% orientation-preserving bijectivity ($\det D\Phi > 0$, zero grid folds). **(C)** Bidirectional invertibility test showing that Reverse ($T \to S$) 600-point fits match or exceed Forward ($S \to T$) fits. **(D)** Unbinned continuous logistic horizon ($t^* = 58.2$ Myr all pairs) and strictly node-disjoint independent taxon pairs ($n = 187$, $t^* = 45.8$ Myr, 54.0% at < 50 Myr vs. 0.0% at ≥ 50 Myr).

Table 4: Causal landmark ablation (9-Head-LM vs. 17-LM vs. 10-Outer-Trunk-LM vs. Direct 600-point silhouette Area of Difference $|\Omega_A \Delta \Omega_B|$), Jacobian diffeomorphic bijectivity ($\min \det D\Phi > 0$), and Reverse-direction ($T \to S$) invertibility across all six of Thompson's 1917 continuous silhouette pairs (600×600 grid).

1917 Pair & Family	Unwarped	9-Head-LM	All 17-LM	10-Trunk-LM	Direct 600-pt	Bijective ($\min \det J$)	Reverse $T \to S$ 600-pt	Reverse IoU
Argyropelecus → *Sternoptyx* (5p Shear)	58.4%	65.2%	63.0%	64.0%	**45.6**%	**100.0**% (1.025)	**57.5**% (vs. 67.4%)	0.545
Scarus → *Pomacanthus* (8p Coaxial)	40.6%	**587.1**%	56.2%	54.5%	**28.5**%	**100.0**% (0.714)	**26.0**% (vs. 40.8%)	**0.765**
Polyprion → *Pristigenys* (8p Fan)	33.2%	45.5%	20.7%	21.0%	**17.2**%	**100.0**% (1.150)	**18.1**% (vs. 42.1%)	**0.834**
Pristigenys → *Scorpaena* (8p Fan)	120.1%	75.0%	59.4%	54.5%	**35.5**%	**100.0**% (0.374)	**35.7**% (vs. 58.7%)	**0.687**
Scorpaena → *Antigonia* (8p Fan)	102.2%	98.0%	143.7%	153.6%	**57.6**%	**100.0**% (0.517)	**39.3**% (vs. 71.0%)	**0.622**
Diodon → *Mola mola* (8p Radial)	70.8%	**8,925**%	69.4%	77.1%	**49.0**%	**100.0**% (0.914)	**32.3**% (vs. 139.3%)	**0.712**

6 Discussion & Causal Synthesis: Reconciling Thompson, Bookstein, and Modern Macroevolution

Why do internal 17-landmark skeletons break global coordinate grids above $\sim$ 45–50 Myr, whereas continuous 600-point silhouettes continue to obey Thompson's 5- to 8-parameter coordinate transformations even across 80–113 Myr interfamily divergences? Our causal ablation and modular residual analyses point to three interlocking mechanisms:

1. **The Statistical Lever-Arm Trap (LM 1–2, 11–17):** Placing 53% of the landmarks inside the anterior $\sim$ 25% of the fish (the skull and opercle) creates a long geometric lever arm: minor trophic remodeling of the jaws and suspensorium rotates or curves the fitted coordinate grid across the posterior 75% of the body, producing up to 587.1% posterior silhouette error on *Scarus* → *Pomacanthus* (and 8,925% on *Diodon* → *Mola*) when head landmarks are fitted alone.

2. **Developmental Modularity (Cranial Neural Crest vs. Somitic Axial Trunk vs. Dermal Fin Rays):** Embryologically, the teleost skull derives from cranial neural crest cells shaped by feeding mechanics, whereas dorsal/anal fin insertions (LM 3–8) migrate along somitic myomeres and distal fin webs grow via dermal lepidotrichia controlled by local bioelectric and morphogen gradients [9, 10]. Even in extreme skeletal apomorphies such as *Mola mola*, where the true caudal fin is lost and replaced by a pseudocaudal *clavus* derived from modified dorsal and anal fin elements (so LM 5–6 mark the posterior clavus boundary), the 8-parameter radial-flare grid captures 79.7% of trunk insertion variance while the hypertrophied dorsal/anal fin span scales 2.26× taller (0.786 vs. 0.349). Within a genus or young family ($\leq$ 45 Myr), global allometric growth gradients scale these modules together (53%–61% noise-floor attainment); beyond the $\sim$ 45–50 Myr horizon, trophic skull evolution and fin-base migration decouple.

3. **Hydrodynamic Selection on the Unified Wetted Silhouette (and Complementary Utility of Both Protocols):** Ichthyologists intentionally standardized the 17-landmark body-wall protocol because museum specimens vary in fin erection and pinning posture [6, 7]. That protocol remains ideal for posture-invariant skeletal and trophic comparisons. However, in living, swimming fishes (and in the standard lateral taxonomic plates with fully deployed median fins that Thompson selected in 1917), water does not flow around an internal 8-vertex skeletal trunk polygon — it flows around the continuous outer silhouette of the body plus its dorsal, anal, and caudal fins acting as a single integrated hydrofoil. Because hydrodynamic drag, added mass, and turning torque act on the *entire lateral contour*, deep-bodied fishes (*Pomacanthus*, *Antigonia*, *Sternoptyx*, *Mola*) co-adapt their dorsal fin, anal fin, and trunk profile as a unified biomechanical envelope — the exact physical field of force that D'Arcy Thompson hypothesized in 1917.

Data and Code Availability

All Python scripts, SHA-256-verified museum specimen landmarks, and full 4,505-pair and silhouette stress-test output tables are openly available in the repository:

- `thompson/1917-thompson-four-families-atlas.py`
- `thompson/1917-thompson-scarus-pomacanthus.py`
- `thompson/1917-thompson-silhouette-area-paradox.py`
- `thompson/run_rigor_stress_tests.py`
- `datasets/thompson/thompson_1917_exact_species_landmarks.json`
- `thompson_taxonomic_scaling_summary.csv`
- `thompson_four_families_summary.json`
- `thompson_silhouette_area_summary.json`
- `thompson_rigor_stress_tests.json`

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